# A Common Ground Consortium assessment for Nuclear Siting Tools


R. A. Borrelli · Denia Djokić

University of Idaho
Department of Nuclear Engineering and Industrial Management

Nuclear Engineering & Radiological Sciences
Fastest Path to Zero Initiative
University of Michigan


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# Collaboration-Based Siting of Spent Nuclear Fuel

# 2021 Consolidated Appropriations Act allocated funds for ‘expenses necessary for nuclear waste disposal activities . . . ’

‘. . . including storage activities’

DOE awarded $26M to thirteen (now twelve) consortia across the nation for building capacity in Collaboration-Based Siting (CBS)

Support more comprehensive community engagement to collaborate with stakeholders to manage SNF effectively

Not actual siting

Process and frameworks

# Twelve consortia awarded across the country

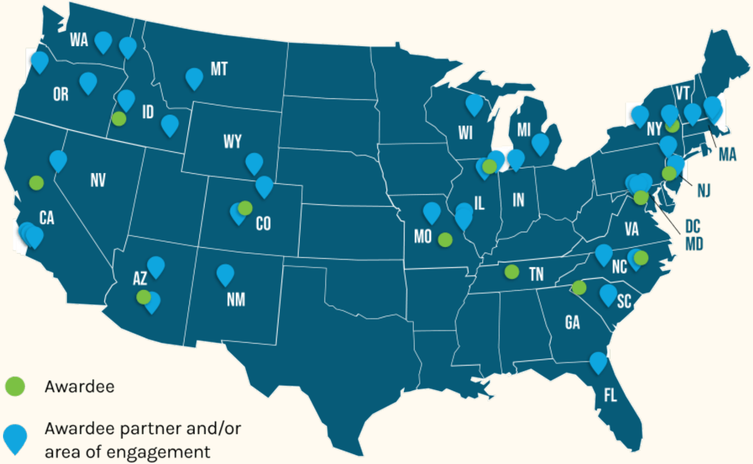


**Figure.** Nationwide consortia

[1] **Department of Energy, 2023. DOE awards $26 million to support consent-based siting for spent nuclear fuel**

# The Common Ground Consortium (CGC)

# CGC spans several institutions across the intermountain west and Denia at the University of Michigan

- With Nation Tribal Energy Association
- Ended in March 2026
- But we got a NEUP!
- Centered on public input
- Research focus on historical lessons, types of public decision-making, and state-specific and/or cross-jurisdictional challenges in Collaboration-Based Siting of critical infrastructure
- Providing recommendations to DOE


[2] Araújo, K., 2026. From Stalemate to Collaboration: Rethinking Nuclear Waste Siting and Policy. Phoenix, Arizona: Proc., Waste Management Symposium

[3] Koerner, C. et al., 2025. The Willrich report predictions. Washington, D. C.: Proc., IHLRWM

[4] Borrelli, R. A., et al., 2024. The Common Ground Consortium Focus on Public Conversations. Las Vegas, Nevada: Proc., ANS Annual Meeting

# Motivation & Goals

# Lots of executive orders were issued in 2024 for nuclear energy growth

400 GWe by 2025!

Deploying reactors at military bases

Recycling & reprocessing

Hopefully pyroprocessing for Christina and me

Increasing domestic fuel production

July 4th milestone

# All this growth really requires strong back end management policy

- More SNF facilities are going to be needed at the least
- Communities or localities may want help in terms of technical feasibility, local capacity to host a site
- There are some tools out there that could provide site screening information
- We critically reviewed and tested some of them within this context
- Based on community accessibility and utility just for an SNF Consolidated Storage Facility (CSF) for now
- Because we were only funded for that at the time
- Subset of a larger study for journal submission
- Just four today

# Siting Tool for Advanced Nuclear Development (STAND)

# STAND examines feasibility for advanced nuclear facility siting

Population, socioeconomics, safety, regulations, etc.

Comparative analysis for different sites

Site Exploration option with economic, proximity, reference, safety data layers

[5] Fastest Path to Zero, 2022. STAND – Quickstart guide. INLRPT-23-74581

# STAND example for Bonneville County siting

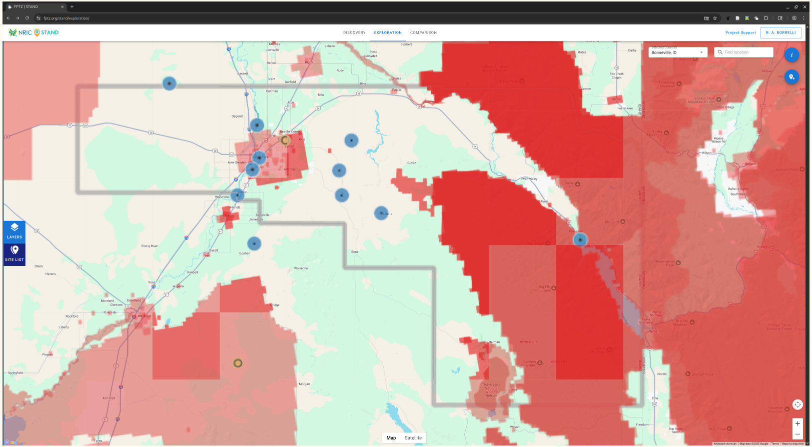

**Figure.** User generated

Red – protected lands

Blue – Electric generators

Yellow – Brownfield site

STAND could be useful for screening

Could use more information on regulations

And more on the criteria themselves

# Web-Based Transportation Routing Analysis Geographic Information System (WebTRAGIS)

# WebTRAGIS is a GIS web tool developed by ORNL for transportation routing

- Databases for highway, rail, waterways
- Select origin and destination to generate route
- Optimize transportation under distance, preferred routes, time
- Layers are county boundaries, critical infrastructure, population, etc.

[6] Peterson, S., 2018. WebTRAGIS: Transportation Routing Analysis Geographic System User's Manual. ORNL/TM-2018/856

# WebTRAGIS national population density with railways

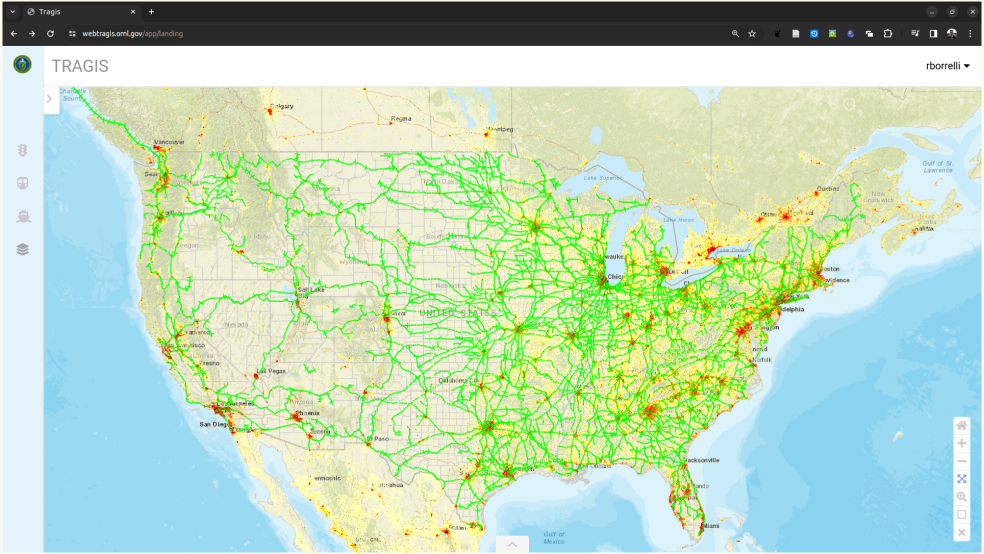


**Figure.** User generated

# WebTRAGIS transport from Richland to Naturita

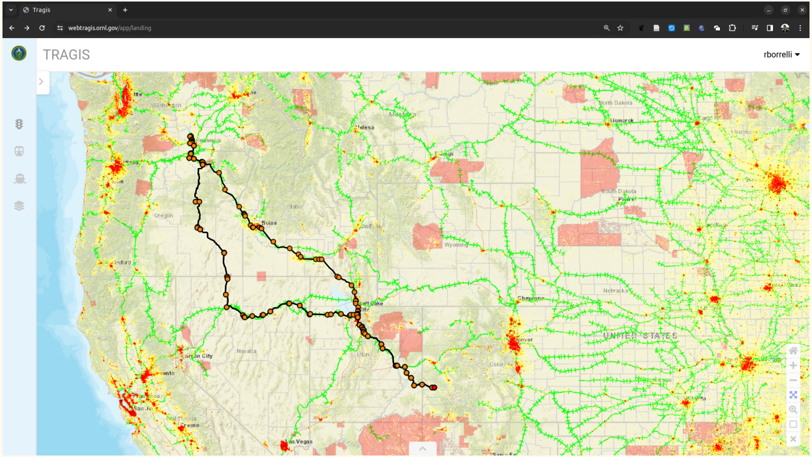


**Figure.** User generated

Southern route includes Idaho

Direct route goes through Idaho

Node based generation

1300 miles & 32 hours

Several data layers

Multiple travel modes

Nodes not shown beforehand; only after destination

Communities might investigate locations of critical infrastructure, population

Communities would not be planning transportation routes though

# Land-area Identification, Tagging, and Exploration (LITE)

# LITE (ORNL) to evaluate areas for storage

Compare up to two locations

Three main areas of functionality –

- Explore – Evaluate areas of interest
- Access – Select assessment considerations
- Compare – Generate assessment reports

[7] Le, T. et al., 2023. Developing digital tools for engagement. Presentation

# LITE site screening criteria for Bonneville county

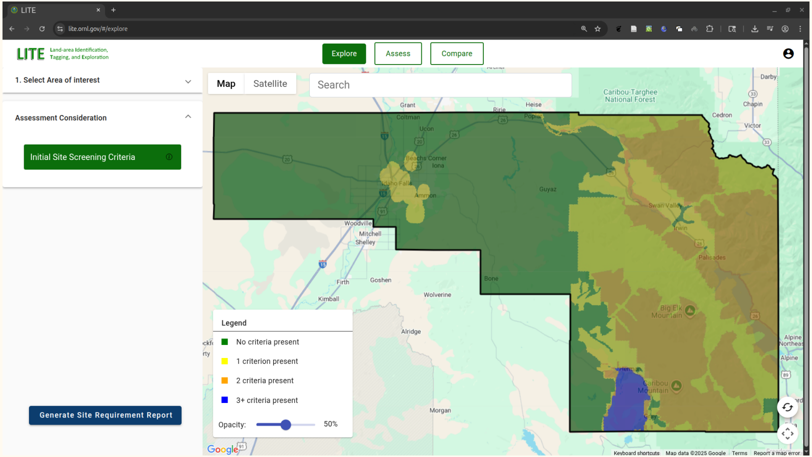


**Figure.** User generated

Layers include Critical Habitat for Endangered and Threatened Species, Coastal Barriers, Federally Designated Lands, etc.

Energy facilities, floodplains, hazardous facilities

West is wide open

East has been flagged under the criteria

# LITE site screening for Bonneville county with all assessment considerations included

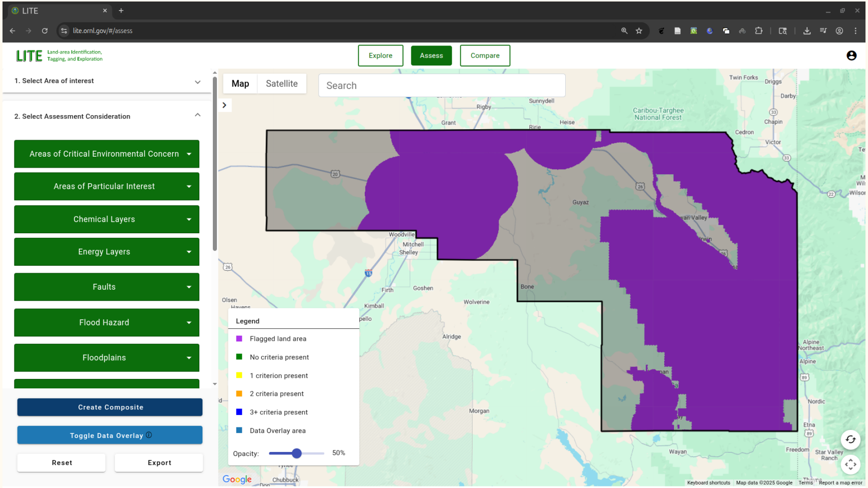


**Figure.** User generated

# LITE comparison for Bonneville county & Southwest Wyoming

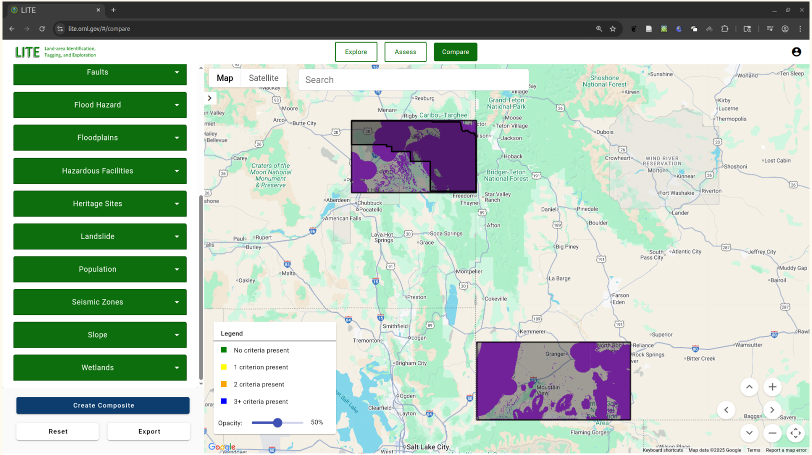


**Figure.** User generated

Tried to select similar area

Same assessment considerations

Wyoming offers more land due to lower population density

LITE offers limited capabilities

Granularity only down to counties

Comparison feature is useful

# Stakeholder Tool for Assessing Radioactive Transportation (START)

# START is a GIS decision tool for SNF transport

Designed by DOE

Generates routes by user selected options; e.g., minimizing distance, population, etc.

Select various sites; e.g., infrastructure, police, airports, casinos, etc.

Select origin, destination, transportation method

Multiple transport modes

**[8] Abkowitz, M. et al., 2016. Application of a Decision-Support Tool for Evaluating Radioactive Material Transportation Routing Options and Emergency Preparedness. Kobe: Proc., 18th International Symposium on the Packaging and Transportation of Radioactive Materials**

# START transport from Richland to Pueblo

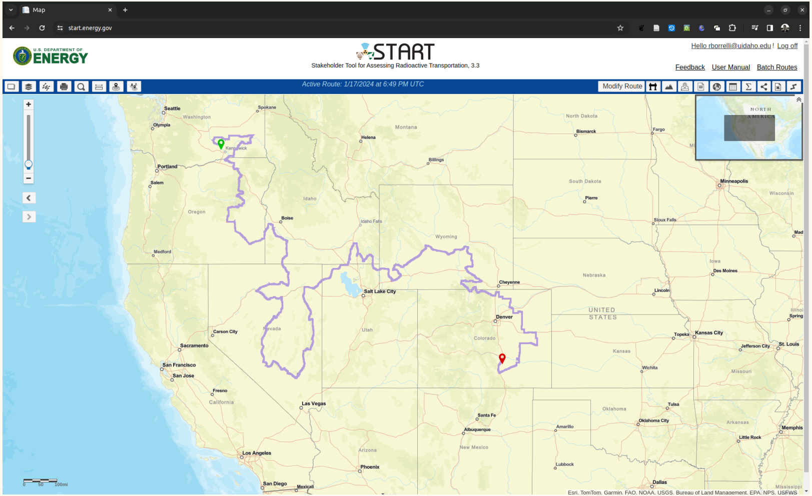


**Figure.** User generated

Heavy haul truck

Minimize population with 800 m buffer

3500 miles

Windy route due to buffer

Not much detail

Difficulty in selecting cities

Could be used to identify critical sites on a route

# Discussion

# These tools have drawbacks

Some tools could not be included due to the change in funding mandate

Could not get access to OR-SAGE – Looked promising from literature

A lot of redundancy across all tools studied – Silos

WebTRAGIS & START are basically the same

WebTRAGIS seemed better with options and detail

Same for STAND over LITE

Major drawback is that the sites require registration

Communities may be wary of privacy issues or if the tool owners may be collecting data without consent

Cannot download the images as jpg, png

Use of the tools in the literature was not prevalent

More transparency on data sources should be provided

Lack of open source accessibility

Would be difficult for public at large use without assistance

# On the other hand, how could we aid in community decisionmaking?

Site screening and transportation routes do offer use if only just to see critical sites

Could be used for public comments

Community could use tools to develop a go/no-protocol in a siting process

Honestly, not that great overall

Easy to see what fixes are needed

# Tools pros & cons I

**Table.** Summary of tools

| Tool | Use | Features | Limits |
|---|---|---|---|
| STAND site | For advanced reactor siting. | Site Exploration methodology could be useful for community planners for screening. Selecting criteria was easy to perform. | Cannot download the map. Criteria lacked sufficient explanations. Registration required. |

# Tools pros & cons II

| | | | |
|---|---|---|---|
| WebTRAGIS site | Modeling transportation routing. Uses GIS to identify optimal means of transportation for user selected parameters. | Intuitive GUI allows selection of a lot of layers including population density and Tribal lands. Can combine transport routes manually. | Cannot download the map as a .jpg, etc. National parks, wilderness not included in layers. Cannot just click on the map to plan route; Transportation 'nodes' are not shown on the map. Registration required. |

# Tools pros & cons III

| | | | |
|---|---|---|---|
| LITE site | Evaluate areas of interest for interim storage using a pre-configured assessment considerations. | Presents a map of screening criteria and assessment considerations down to the county level. Areas can also be compared. Descriptions of the criteria and considerations is provided. | Cannot download the map as a .jpg., etc. Lacking granularity. Registration required. |
| START site | Geospatial decision-support tool developed for evaluating routing options and other aspects of transporting SNF. | Variety of data layers. Printable map options. Can select different transportation modes for one shipping route. Can select for distance, time, population buffer. Can select a wide range of infrastructure to filter. | Hard to select cities, etc., for destinations. Have to zoom in significantly to select a destination. Map renders slowly. |

# Wrapping up

# What is next?

- There are a few more tools to check out
- Propose a methodology for Collaboration-Based Siting incorporating some of the tools
- For effective back-end management, community autonomy can be prioritized by providing the necessary tools for informed, independent decision-making
- Finish the journal paper

# References


1. Department of Energy, 2023. DOE awards $26 million to support consent-based siting for spent nuclear fuel.
2. Araújo, K., 2026. From Stalemate to Collaboration: Rethinking Nuclear Waste Siting and Policy. Phoenix, Arizona: Proc., Waste Management Symposium.
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6. Peterson, S., 2018. WebTRAGIS: Transportation Routing Analysis Geographic System User's Manual. ORNL/TM-2018/856.
7. Le, T. et al., 2023. Developing digital tools for engagement. Presentation.
8. Abkowitz, M. et al., 2016. Application of a Decision-Support Tool for Evaluating Radioactive Material Transportation Routing Options and Emergency Preparedness. Kobe: Proc., 18th International Symposium on the Packaging and Transportation of Radioactive Materials.

# Acronyms I

**ANS** American Nuclear Society.

**CAA** Consolidated Appropriations Act.

**CBS** Collaboration-Based Siting.

**CGC** Common Ground Consortium.

**CSF** Consolidated Storage Facility.

**DOE** United States Department of Energy.

**GIS** Geographic Information System.

**GUI** Graphical User Interface.

**GWe** Gigawatts-electric.

**LITE** Land-area Identification, Tagging, and Exploration.

**NEUP** Nuclear Energy University Program.

# Acronyms II

**OR-SAGE** Oak Ridge Siting Analysis for Power Generation Expansion.

**ORNL** Oak Ridge National Laboratory.

**SNF** Spent Nuclear Fuel.

**STAND** Siting Tool for Advanced Nuclear Development.

**START** Stakeholder Tool for Assessing Radioactive Transportation.

**UM** University of Michigan.

**WebTRAGIS** Web-Based Transportation Routing Analysis Geographic Information System.